

LIFEWS AgriShine(TM) GrowMeal(TM)

SSS 1-3 ASSESSMENT & QUIZ SYSTEM

STUDENT ASSESSMENT BOOKLET

This booklet assesses four dimensions: knowledge, practical competence, investigation/data skills, and mission-based problem solving.

Knowledge	15%
Practical competence	20%
Investigation/Data	30%
Mission problem solving	35%

Name: _____ Class: _____

School: _____ Term/Date: _____

SECTION A - KNOWLEDGE

AQ-401 Frame a research or engineering question involving yield components, harvest index and resource-use efficiency.

AQ-402 Identify one assumption and one constraint relevant to the advanced production plot.

AQ-403 Explain why validation is distinct from calibration or fitting.

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-404: Execute or demonstrate a technically appropriate measurement of efficiency metric, with instrument ID, unit and QC note.		/3
AQ-405: Document one calibration, verification or reference check.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-406 Analyze a dataset using an appropriate quantitative metric and state uncertainty.

AQ-407 Compare at least two scenarios/designs and quantify the performance trade-off.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-408 Stress-test the conclusion by changing one assumption or input and report whether the decision changes.

AQ-409 Recommend an optimization and define a measurable acceptance criterion.

AQ-410 Write a decision statement integrating performance, uncertainty, safety and implementation constraints.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-411 Frame a research or engineering question involving replicated treatments, blocks, variability and treatment effects.

AQ-412 Identify one assumption and one constraint relevant to the crop trial.

AQ-413 Explain why validation is distinct from calibration or fitting.

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-414: Execute or demonstrate a technically appropriate measurement of treatment effect, with instrument ID, unit and QC note.		/3
AQ-415: Document one calibration, verification or reference check.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-416 Analyze a dataset using an appropriate quantitative metric and state uncertainty.

AQ-417 Compare at least two scenarios/designs and quantify the performance trade-off.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-418 Stress-test the conclusion by changing one assumption or input and report whether the decision changes.

AQ-419 Recommend an optimization and define a measurable acceptance criterion.

AQ-420 Write a decision statement integrating performance, uncertainty, safety and implementation constraints.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-421 Frame a research or engineering question involving causal variables, factorial effects, confounding and sensitivity.

AQ-422 Identify one assumption and one constraint relevant to the controlled experiment.

AQ-423 Explain why validation is distinct from calibration or fitting.

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-424: Execute or demonstrate a technically appropriate measurement of experimental metric, with instrument ID, unit and QC note.		/3
AQ-425: Document one calibration, verification or reference check.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-426 Analyze a dataset using an appropriate quantitative metric and state uncertainty.

AQ-427 Compare at least two scenarios/designs and quantify the performance trade-off.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-428 Stress-test the conclusion by changing one assumption or input and report whether the decision changes.

AQ-429 Recommend an optimization and define a measurable acceptance criterion.

AQ-430 Write a decision statement integrating performance, uncertainty, safety and implementation constraints.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-431 Frame a research or engineering question involving flow, head, pipe loss, uniformity and pumping efficiency.

AQ-432 Identify one assumption and one constraint relevant to the irrigation system.

AQ-433 Explain why validation is distinct from calibration or fitting.

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-434: Execute or demonstrate a technically appropriate measurement of engineering metric, with instrument ID, unit and QC note.		/3
AQ-435: Document one calibration, verification or reference check.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-436 Analyze a dataset using an appropriate quantitative metric and state uncertainty.

AQ-437 Compare at least two scenarios/designs and quantify the performance trade-off.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-438 Stress-test the conclusion by changing one assumption or input and report whether the decision changes.

AQ-439 Recommend an optimization and define a measurable acceptance criterion.

AQ-440 Write a decision statement integrating performance, uncertainty, safety and implementation constraints.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-441 Frame a research or engineering question involving water balance, ET, IWP and ET water productivity.

AQ-442 Identify one assumption and one constraint relevant to the water-productivity dataset.

AQ-443 Explain why validation is distinct from calibration or fitting.

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-444: Execute or demonstrate a technically appropriate measurement of water-productivity metric, with instrument ID, unit and QC note.		/3
AQ-445: Document one calibration, verification or reference check.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-446 Analyze a dataset using an appropriate quantitative metric and state uncertainty.

AQ-447 Compare at least two scenarios/designs and quantify the performance trade-off.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-448 Stress-test the conclusion by changing one assumption or input and report whether the decision changes.

AQ-449 Recommend an optimization and define a measurable acceptance criterion.

AQ-450 Write a decision statement integrating performance, uncertainty, safety and implementation constraints.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-451 Frame a research or engineering question involving pH/EC dynamics, flow and residence time.

AQ-452 Identify one assumption and one constraint relevant to the advanced soilless unit.

AQ-453 Explain why validation is distinct from calibration or fitting.

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-454: Execute or demonstrate a technically appropriate measurement of soilless metric, with instrument ID, unit and QC note.		/3
AQ-455: Document one calibration, verification or reference check.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-456 Analyze a dataset using an appropriate quantitative metric and state uncertainty.

AQ-457 Compare at least two scenarios/designs and quantify the performance trade-off.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-458 Stress-test the conclusion by changing one assumption or input and report whether the decision changes.

AQ-459 Recommend an optimization and define a measurable acceptance criterion.

AQ-460 Write a decision statement integrating performance, uncertainty, safety and implementation constraints.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-461 Frame a research or engineering question involving accuracy, precision, calibration, MAE/RMSE and network design.

AQ-462 Identify one assumption and one constraint relevant to the monitoring station.

AQ-463 Explain why validation is distinct from calibration or fitting.

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-464: Execute or demonstrate a technically appropriate measurement of monitoring metric, with instrument ID, unit and QC note.		/3
AQ-465: Document one calibration, verification or reference check.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-466 Analyze a dataset using an appropriate quantitative metric and state uncertainty.

AQ-467 Compare at least two scenarios/designs and quantify the performance trade-off.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-468 Stress-test the conclusion by changing one assumption or input and report whether the decision changes.

AQ-469 Recommend an optimization and define a measurable acceptance criterion.

AQ-470 Write a decision statement integrating performance, uncertainty, safety and implementation constraints.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-471 Frame a research or engineering question involving thresholds, hysteresis, feedback and adaptive control.

AQ-472 Identify one assumption and one constraint relevant to the smart irrigation controller.

AQ-473 Explain why validation is distinct from calibration or fitting.

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-474: Execute or demonstrate a technically appropriate measurement of control metric, with instrument ID, unit and QC note.		/3
AQ-475: Document one calibration, verification or reference check.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-476 Analyze a dataset using an appropriate quantitative metric and state uncertainty.

AQ-477 Compare at least two scenarios/designs and quantify the performance trade-off.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-478 Stress-test the conclusion by changing one assumption or input and report whether the decision changes.

AQ-479 Recommend an optimization and define a measurable acceptance criterion.

AQ-480 Write a decision statement integrating performance, uncertainty, safety and implementation constraints.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-481 Frame a research or engineering question involving problem definition, requirements, concept selection and prototype testing.

AQ-482 Identify one assumption and one constraint relevant to the design brief.

AQ-483 Explain why validation is distinct from calibration or fitting.

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-484: Execute or demonstrate a technically appropriate measurement of design metric, with instrument ID, unit and QC note.		/3
AQ-485: Document one calibration, verification or reference check.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-486 Analyze a dataset using an appropriate quantitative metric and state uncertainty.

AQ-487 Compare at least two scenarios/designs and quantify the performance trade-off.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-488 Stress-test the conclusion by changing one assumption or input and report whether the decision changes.

AQ-489 Recommend an optimization and define a measurable acceptance criterion.

AQ-490 Write a decision statement integrating performance, uncertainty, safety and implementation constraints.

Student reflection: What evidence from this assessment are you most confident about?

SECTION A - KNOWLEDGE

AQ-491 Frame a research or engineering question involving system boundaries, balances, time-varying mismatch, cascades and scenarios.

AQ-492 Identify one assumption and one constraint relevant to the integrated system model.

AQ-493 Explain why validation is distinct from calibration or fitting.

SECTION B - PRACTICAL COMPETENCE

Task	Student evidence	Score
AQ-494: Execute or demonstrate a technically appropriate measurement of integrated-system metric, with instrument ID, unit and QC note.		/3
AQ-495: Document one calibration, verification or reference check.		/3

SECTION C - INVESTIGATION / DATA SKILLS

AQ-496 Analyze a dataset using an appropriate quantitative metric and state uncertainty.

AQ-497 Compare at least two scenarios/designs and quantify the performance trade-off.

SECTION D - MISSION-BASED PROBLEM SOLVING

AQ-498 Stress-test the conclusion by changing one assumption or input and report whether the decision changes.

AQ-499 Recommend an optimization and define a measurable acceptance criterion.

AQ-500 Write a decision statement integrating performance, uncertainty, safety and implementation constraints.

Student reflection: What evidence from this assessment are you most confident about?

MY GROWMEAL(TM) ASSESSMENT SUMMARY

My strongest area: _____

One skill I will improve: _____

One mission I want to try again: _____

Teacher feedback:
